

MP1994V2 Milestone Tracker

Mortensen–Pissarides (1994) Lean replication

29 July 2026

Last completed milestone: M1 – Surplus Bellman equation – **COMPLETE – GREEN**

Next milestone: M2 – Affine surplus and endogenous reservation cutoff – **NOT STARTED**

Source of truth. docs/milestone_tracker.md. Status vocabulary: NOT STARTED; IN PROGRESS; BLOCKED; READY FOR REVIEW; COMPLETE – GREEN; COMPLETE – AMBER; DEFERRED / OUTSIDE CORE LEAN. M0 and M1 are complete and green. M2 has not started.

Milestone plan

The tracker uses one readable block per milestone instead of a dense landscape table. Each block contains the same fields as the Markdown table.

M0. Foundations and primitive value equilibrium

Status: **COMPLETE – GREEN**

Paper: Sections 2–3; equations (1)–(7)

Economic target: Primitives, layered assumptions, probability/matching definitions, and primitive value equilibrium.

Expected declarations: Primitives; four assumption bundles; cdf; meeting rates; surplus; continuationTerm; ValueEquilibrium

Adequacy grade: GREEN **Dependencies:** None

Blocker: None

Branch/commit or tag; last review; next action: issue2; completion commit; 2026-07-29 human mathematical/economic review; preserve the reviewed M0 interface while beginning M1 separately.

M1. Surplus Bellman equation

Status: **COMPLETE – GREEN**

Paper: Section 3, journal p. 401; equation (8)

Economic target: Derive equation (8) from equations (3)–(7).

Expected declarations: Continuation decomposition; firm_share; surplus_flow_equation; surplus_bellman_of_probability; surplus_bellman

Adequacy grade: GREEN **Dependencies:** M0 complete

Blocker: None

Branch/commit or tag; last review; next action: issue2; completion commit; 2026-07-29 human mathematical/economic review; preserve the reviewed conditional representation theorem.

M2. Affine surplus and endogenous reservation cutoff

Status: **NOT STARTED**

Paper: Section 3; reservation discussion and equation (12)

Economic target: Surplus difference theorem, strict monotonicity, unique endogenous reservation threshold, and equation (12).

Expected declarations: Surplus difference, affinity, strict monotonicity, zero uniqueness, reservation theorem

Adequacy grade: – **Dependencies:** M1

Blocker: None

Branch/commit or tag; last review; next action: No branch/tag or review. The two-point surplus identity and strict monotonicity follow from equation (8) under $r + \lambda > 0$ and $\sigma > 0$. Economic cutoff admissibility, especially $\varepsilon_d < \varepsilon_u$, must be proved separately using equations (1)–(2), vacancy-cost and meeting-rate positivity, and $\beta < 1$.

M3. Continuation value, job destruction, and job creation Paper: Section 3; equations (9), (10), and (13) Economic target: Measure-form continuation identity and derivation of job destruction and job creation. Expected declarations: Tail/positive-part identity; derived JD and JC theorems Adequacy grade: – Dependencies: M2 Blocker: Measure-theoretic tail identity Branch/commit or tag; last review; next action: No branch/tag or review; specify atomless support and integrability hypotheses precisely.	Status: NOT STARTED
M4. Reduced-equilibrium reconstruction and equivalence Paper: Section 3 Economic target: Prove value equilibrium to reduced equilibrium and construct a value equilibrium from a reduced solution. Expected declarations: ReducedEquilibrium; forward map; converse constructor; equivalence Adequacy grade: – Dependencies: M3 Blocker: Value reconstruction design Branch/commit or tag; last review; next action: No branch/tag or review; define reduced objects only after JD and JC are theorems.	Status: NOT STARTED
M5. Existence and uniqueness of static equilibrium Paper: Section 3; Figure 1 and statement following equation (13) Economic target: Clarified existence theorem and uniqueness of (θ, ε_d) . Expected declarations: Existence; at-most-one; unique-equilibrium theorem with endpoint assumptions Adequacy grade: – Dependencies: M4 Blocker: Paper omits endpoint conditions Branch/commit or tag; last review; next action: No branch/tag or review; state corrected continuity, range, and crossing hypotheses.	Status: NOT STARTED
M6. Unemployment flow balance Paper: Section 3; equation (14) Economic target: Derive steady-state unemployment and define the full steady state. Expected declarations: Flow balance; unemployment identity; full steady-state structure Adequacy grade: – Dependencies: M4–M5 Blocker: Static equilibrium foundation Branch/commit or tag; last review; next action: No branch/tag or review; introduce unemployment stocks only here.	Status: NOT STARTED
M7. Global comparative statics Paper: Section 3 Economic target: Order-theoretic comparative statics for p, b, λ, r, σ where possible without differentiability. Expected declarations: Parameter-order theorems; equilibrium order lemmas Adequacy grade: – Dependencies: M5 Blocker: Global solution selection and parameter monotonicity Branch/commit or tag; last review; next action: No branch/tag or review; separate global monotone results from derivatives.	Status: NOT STARTED
M8. Appendix derivative comparative statics Paper: Appendix; equations (A1)–(A12) Economic target: Differentiable equilibrium paths and derivative sign results. Expected declarations: Analytic assumption layer; Jacobian/implicit-path theorems Adequacy grade: – Dependencies: M5, M7 Blocker: Differentiability and nondegenerate Jacobian Branch/commit or tag; last review; next action: No branch/tag or review; keep Appendix assumptions out of the core.	Status: NOT STARTED

M9. Two-state anticipated cyclical shocks**Status: NOT STARTED****Paper:** Section 4; equations (15)–(31)**Economic target:** Two-state surplus systems, cutoff ordering under explicit hypotheses, and impact asymmetry.**Expected declarations:** Two-state value system; ordering theorems; employment-measure impact results**Adequacy grade:** – **Dependencies:** M3–M6**Blocker:** Piecewise continuation and employment measure**Branch/commit or tag; last review; next action:** No branch/tag or review; begin with an explicit two-state process.**M10. Finite Markov equilibrium and employment accounting****Status: NOT STARTED****Paper:** Analytical prelude to Section 5; equations (32)–(38)**Economic target:** Finite-state equilibrium representation, employment transition operator, creation/destruction accounting, and equation (38).**Expected declarations:** Finite kernel; transition operator; mass preservation; accounting identity**Adequacy grade:** – **Dependencies:** M6, M9**Blocker:** Finite-state interface and numerical-solver connection**Branch/commit or tag; last review; next action:** No branch/tag or review; prefer finite state before general measurable states.**NUM. Numerical simulation and calibration****Status: DEFERRED / OUTSIDE CORE LEAN****Paper:** Section 5; equations (39)–(42), Tables I–II**Economic target:** External Python or Julia numerical replication; Lean only for exact accounting or certified residuals where useful.**Expected declarations:** External solver and optional Lean residual/accounting certificates**Adequacy grade:** – **Dependencies:** Analytical interfaces as needed**Blocker:** Numerical/empirical work is outside core Lean**Branch/commit or tag; last review; next action:** No branch/tag or review; define a reproducible calibration and residual protocol externally.**M0 acceptance checklist**

- ☒ New module root is isolated from the legacy architecture.
- ☒ Primitive parameter structure exists.
- ☒ Economic assumptions are layered.
- ☒ Shock law is represented by a measure.
- ☒ CDF is derived from the measure.
- ☒ Matching rates are defined.
- ☒ Surplus is defined from J, W, U , not stored independently.
- ☒ Equations (1)–(7) are represented faithfully.
- ☒ Equations (5) and (6) use one shared continuation term.
- ☒ No cutoff or reduced-equilibrium conclusion is assumed.
- ☒ Targeted builds pass.
- ☒ Full build passes.
- ☒ No `sorry`, `admit`, or custom axioms exist.
- ☒ Tracker TeX compiles and PDF has been visually inspected.
- ☒ Human mathematical/economic review is complete.

M1 acceptance checklist

- ☒ Generic continuation decomposition is proved under a probability measure.
- ☒ Firm surplus share is derived from equations (3) and (4).
- ☒ The surplus flow equation is derived from equations (5)–(7).
- ☒ Equation (8) is proved for the actual equilibrium surplus.
- ☒ The main theorem uses only probability normalization and existing integrability.
- ☒ The `ShockAssumptions` wrapper installs the probability instance locally.
- ☒ No M0 structure was strengthened or modified.
- ☒ No cutoff, affine surplus, or reduced equilibrium was introduced.
- ☒ Targeted builds pass.
- ☒ Full build passes.
- ☒ No `sorry`, `admit`, or custom axioms exist.
- ☒ Proof ledger TeX/PDF compiles and passes visual inspection.
- ☒ Milestone tracker TeX/PDF compiles and passes visual inspection.
- ☒ Human mathematical/economic review is complete.

Changelog

2026-07-29 – M1 review completed. Human mathematical and economic review found equation (8) mathematically faithful to the paper and its proof non-circular. It is derived from equations (3)–(7), uses the actual equilibrium surplus, and assumes no cutoff, affinity result, or reduced equilibrium. It is a conditional representation theorem for an existing `ValueEquilibrium`; M1 does not prove equilibrium existence. M1 was graded **COMPLETE – GREEN**.

2026-07-29 – M1 equation (8) implemented. Created `MP1994V2/SteadyState/Surplus.lean` with the continuation decomposition, firm share, surplus flow equation, minimal probability theorem, and `ShockAssumptions` wrapper. Updated `All`, `Audit`, `README`, trackers, and added the proof ledger. M1 is **READY FOR REVIEW**.

2026-07-29 – M0 review completed. Review found equations (1)–(7) economically faithful and no future conclusion embedded; corrected substantive modules $\rightarrow$ `All` $\rightarrow$ `Audit`; required explicit `ShockAssumptions P` and local `D.isProbability` in M1; graded **COMPLETE – GREEN**.

2026-07-29 – v2 foundation created. Added `Definitions`, `Assumptions`, `Probability`, `Matching`, `Equilibrium/Value`, `All`, `Audit`, `README`, and synchronized trackers: primitives, four assumption bundles, CDF/support/matching notation, shared surplus/continuation, `ValueCandidate`, and `ValueEquilibrium`. No legacy v1 Lean or architecture file changed; the library-root import makes `lake build` certify `MP1994V2.All`.